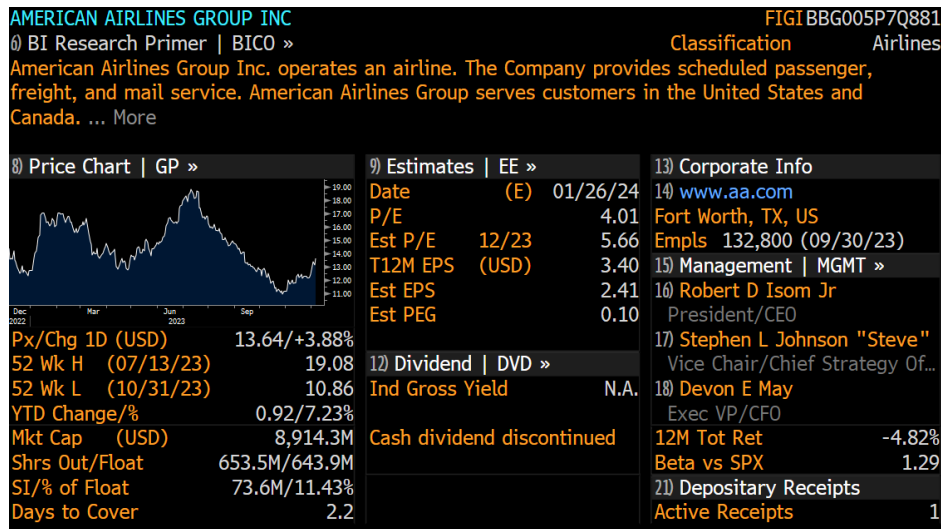


Investments Homework Assignment

Dynamic Hedging - a Discrete Approach

1. Choose any single stock that is a member of the S&P 500 Index



The company we chosen is **AMERICAN AIRLINES GROUP INC.**

2. Choose any OTM strike for a call option on the stock, expiry 15th December 2023

The share price of American airlines on 10th November 2023 is \$11.795, so we chose call option with a strike price \$12.

3. Number of contracts = $10 + (KM1 + KM2) / 100$

KM1	7201
KM2	7242.12
Number of contracts	155
Total notional	15500

4. Option price and option premium

From the image below (data source: Bloomberg), we can see that the option price for one share is \$0.4786, and the total option premium is $15500 * 0.4786 = \$7417.54$

Option pricing			
American Vanilla			
Underlying	AAL US Equity		
Und. Price	USD	Mid	11.795
Trade		11/10/2023	23:30
Settle		11/10/2023	
Style		Vanilla	American
Call/Put		Call	
Direction / Position		Sell	15,500.00
Strike	% Money	1.74% OTM	12.00
Expiry		12/15/2023	23:30
Time to Expiry		35	00:00
Model		BS - disc.	
Vol	BVOL	Bid	37.557%
More Market Data			
Greeks			
Delta (%)			-47.98
Gamma (%)			-3.4259
Vega			-225.56
Theta			122.28
Rho			-0.77
Time Value			-7,417.54
Gearing			-24.65
Break-Even (%)			5.80
Advanced Greeks			
Results			
Price (Total)	USD		-7,417.54
Price (Share)			-0.4786
Price (%)			-4.0572
Margin	Total		0.00

5. To hedge the delta risk of the short call position you buy some stocks at the closing price of 10th November 2023. How many stocks should you buy?

To hedge the delta risk of short call we must buy $15500 * |\Delta| = 15500 * 47.98 = 743690$ shares of American Airlines.

6. Until expiry at the end of each day you calculate your total delta (delta of the short call + delta of the stocks), and you buy or sell some stocks in an amount that would make your total delta close to zero. Show the list of the required stock trades (stock amount, buy/sell, stock price, date).

Chen Hang (D1CDZC)
Gao Weijun (EVMNP2)

Start Date	11/10/2023							
End Date	12/15/2023							
Shares	15500							
Strike Price	12							
	AAL US Equity							
Dates	MID_PRICE	DELTA	TRADED VOLUME	CUMULATE D TRADED VOLUME	TRADED	DIRECTION	COST	CUMULATIVE COST
11/10/2023	11.795	-47.98%	7436.90	7436.90	7436.9	buy	-87718.24	-87718.24
11/13/2023	11.775	-46.87%	-172.05	7264.85	7608.95	sell	2025.89	-85692.35
11/14/2023	12.235	-60.96%	2183.95	9448.80	9792.9	buy	-26720.63	-112412.98
11/15/2023	12.435	-66.56%	868.00	10316.80	10660.9	buy	-10793.58	-123206.56
11/16/2023	12.205	-59.87%	-1036.95	9279.85	11697.85	sell	12655.97	-110550.58
11/17/2023	12.285	-62.60%	423.15	9703.00	12121	buy	-5198.40	-115748.98
11/20/2023	12.395	-66.85%	658.75	10361.75	12779.75	buy	-8165.21	-123914.18
11/21/2023	12.135	-58.24%	-1334.55	9027.20	14114.3	sell	16194.76	-107719.42
11/22/2023	12.305	-63.86%	871.10	9898.30	14985.4	buy	-10718.89	-118438.31
11/24/2023	12.315	-64.89%	159.65	10057.95	15145.05	buy	-1966.09	-120404.40
11/27/2023	12.185	-60.39%	-697.50	9360.45	15842.55	sell	8499.04	-111905.36
11/28/2023	12.155	-59.37%	-158.10	9202.35	16000.65	sell	1921.71	-109983.65
11/29/2023	12.235	-62.77%	527.00	9729.35	16527.65	buy	-6447.85	-116431.50
11/30/2023	12.445	-71.24%	1312.85	11042.20	17840.5	buy	-16338.42	-132769.92
12/1/2023	13.035	-86.82%	2414.90	13457.10	20255.4	buy	-31478.22	-164248.14
12/4/2023	13.345	-91.99%	801.35	14258.45	21056.75	buy	-10694.02	-174942.15
12/5/2023	13.135	-90.30%	-261.95	13996.50	21318.7	sell	3440.71	-171501.44
12/6/2023	13.475	-95.59%	819.95	14816.45	22138.65	buy	-11048.83	-182550.27
12/7/2023	13.925	-97.95%	365.80	15182.25	22504.45	buy	-5093.77	-187644.03
12/8/2023	13.775	-97.94%	-1.55	15180.70	22506	sell	21.35	-187622.68
12/11/2023	13.725	-98.59%	100.75	15281.45	22606.75	buy	-1382.79	-189005.47
12/12/2023	14.035	-98.74%	23.25	15304.70	22630	buy	-326.31	-189331.79
12/13/2023	14.095	-99.41%	103.85	15408.55	22733.85	buy	-1463.77	-190795.55
12/14/2023	14.615	-100%	91.45	15500.00	22825.3	buy	-1336.54	-192132.09
12/15/2023	14.495	0.00%	-15500.00	0.00	38325.3	sell	186000.00	-6132.09

- 7. How many stocks have you traded altogether during the lifetime of your option position? What is the ratio of total stock traded compared to the notional of the option?**

The sum of the absolute number of the traded stocks: 38325.3 shares

Ratio of total stock traded compared to the notional of the option: $38325.3 / 15500 = 2.4726$

- 8. After expiry you will have zero option position and you should have Zero stock position since you will hedge back all your delta risk to zero. However, you probably will have some profit or loss after executing this strategy. What is the result of your strategy? why is it not zero?**

Chen Hang (D1CDZC)
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Result: The total profit is 1285.45 dollars.

Option Premium	7417.54
Hedge Cash Flow	-6132.09
Net Profit	1285.45

We have profit because at the time we sell the option, the volatility of the stock price is high, so we sell it with a good price and during the lifetime of the option, we are lucky because of smaller volatility which makes the change in the stock price small, So, the accumulated cost of hedging is \$-6132.09, but we have an option premium of \$7417.54, therefore, the net profit is 1285.45, which is not 0.